SHREENIJA VADAYAR

Design & Manufacturing Engineer | Prototyping | Quality & Process Improvement

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SUMMARY

Entry-level engineer with over a year of hands-on experience at Tecomet in medical-device component development, working alongside Engineering, Manufacturing, and Quality teams. Pairs CAD and documentation discipline with hands-on prototyping, taking ideas from first model through fabrication, testing, and iteration. Seeking design engineering, manufacturing engineering, and product development roles.

EDUCATION

University of Colorado Boulder *May 2025*

B.S. Creative Technology & Design. Minors: Business & Entrepreneurship; Leadership Studies

EXPERIENCE

Tecomet: Engineering Intern / Co-Op *Boulder, CO | June 2022 – July 2023*

- Supported a process-improvement initiative on recurring dimensional and visual defects by expanding inspection samples, collecting and analyzing data, and working with Engineering, Manufacturing, and Quality to evaluate causes and solutions.
- Contributed to implementing a revised production-lot size that substantially reduced defects, then monitored subsequent lots for about two weeks to help confirm the improvement was sustained.
- Performed visual and dimensional inspections of manufactured components, evaluating surface defects and measurements against engineering drawings, specifications, and tolerances.
- Documented nonconformances and communicated recurring defects to departmental leads, contributing inspection findings and production data to cross-functional investigations.
- Translated evolving client requirements into revisions of existing SolidWorks models, engineering drawings, BOMs, and Process Control Plans, maintaining 50+ documents that supported 20–25+ parts orders across 2–3 concurrent workstreams.
- Coordinated documentation and design updates across Engineering, Manufacturing, and Quality to keep client specifications, production requirements, and deliverables aligned; prepared for client meetings, recorded decisions, and followed up on action items.

ATLAS Institute, University of Colorado Boulder: Undergraduate Research Assistant *January 2023 – May 2023*

- Evaluated fabrication methods and material options for a graduate research project through hands-on prototyping, structured testing, and comparative analysis.
- Produced and evaluated 20+ prototype samples and trials using 3D printing, laser cutting, and alternative materials, including filaments and textiles for component casing and cushioning.
- Used SketchUp and SolidWorks for 3D modeling and revisions, assessing fabrication quality, dimensional precision, production time, and material behavior.
- Recorded observations and analyzed results in Excel and through manual review to compare prototype performance and fabrication outcomes.
- Communicated findings to undergraduate and graduate researchers, informing material and fabrication decisions for a final prototype prepared for a funding presentation.

PROJECTS

Woven Voices: Interactive Wearable Technology Installation *Senior Capstone, Solo | Sept 2024 – May 2025*

- Designed and fabricated a wearable installation of seven mechanical flowers that respond to visitor proximity, combining garment construction, brass-rod mechanisms, and 3D-printed housings.
- Integrated micro servos, VL53L0X distance sensors, Adafruit Trinket M0 boards, and NeoPixel LEDs, and wrote Arduino programs that adjust servo movement and LED brightness as visitors approach, along with additional garment lighting effects.
- Iteratively tested and troubleshot servo mounting, structural rigidity, wiring, power, and system integration.
- Presented at the 2025 CU Boulder Design Expo to 25+ visitors; awarded First Prize in the Autodesk Make it Wearable Engineering Student Contest.

Tailgating Detection Device: Physical Computing & Object Design *University of Colorado Boulder | Individual*

- Independently designed, programmed, and fabricated a working prototype that triggers visual and audible warnings for a vehicle following too closely, using an Arduino Uno, HC-SR04 ultrasonic sensor, piezo buzzer, and NeoPixel LEDs.
- Designed the enclosure artwork in Adobe Illustrator and produced the laser-cut housing.
- Programmed and tested sensor-triggered alerts and scrolling LED messages; testing identified the ultrasonic sensor's forward-facing detection range as a design constraint.

Guardian Angel: Wearable Safety Technology *University of Colorado Boulder | Team of four*

- Designed the mechanical form of a wearable safety button for a connected safety concept.
- Contributed to electronics integration and Arduino programming, tested Bluetooth connectivity between the device and its companion experience, and developed elements of the companion interface in Figma.

SoleMate: Wearable Running Analysis Concept *University of Colorado Boulder | Team of five*

- As primary engineering contributor, researched pressure-sensor placement and quantity, sole configuration, and cost for a pressure-sensing insole concept intended to give runners feedback on their mechanics.
- Conducted user interviews, surveys, and competitor and market research to inform product configuration and technical trade-offs. Concept stage; no prototype was built.

MossTak: Sustainable Living Roof Concept *University of Colorado Boulder | Team project*

- Created a SketchUp model and 3D-printed tile representing a moss-based roofing concept, and conducted market and sustainability research.
- Developed Figma UX/UI flows and contributed to website and pitch materials. Concept stage; moss growth and roofing performance were not tested.

Virtual Reality Drawing Environment *University of Colorado Boulder | Individual*

- Independently built an interactive environment from scratch in Unity for a Meta VR headset.
- Implemented drawing and erasing tools, palette and color-wheel color selection, and drawable surfaces including mannequins and a drawing board.

SKILLS

CAD & documentation: SolidWorks, SketchUp, engineering drawings, BOMs, Process Control Plans, technical documentation

Prototyping & fabrication: 3D printing, laser cutting, file preparation, assembly, soldering, wiring, materials evaluation

Electronics & physical computing: Arduino IDE, Arduino Uno, Adafruit Trinket M0, VL53L0X and HC-SR04 sensors, micro servos, NeoPixel LEDs, piezo buzzers, Bluetooth

Programming & interactive technology: Arduino programming, Python, Unity, virtual reality development

Manufacturing & quality: Visual and dimensional inspection, tolerance verification, nonconformance documentation, process control, defect investigation, root-cause analysis, continuous improvement

Design & productivity tools: Adobe Illustrator, Photoshop, Figma, Microsoft Excel and PowerPoint, Google Workspace

CERTIFICATIONS

Lean Six Sigma Green Belt *July 2026*
Council for Six Sigma Certification (CSSC)

Women's Startup Program: Management & Entrepreneurship Certificate
IIM Bangalore (NSRCEL) & Goldman Sachs. Entrepreneurship, product validation, business planning, pitching, and business development